

Neuro-architectural stress briefing

A science-literacy document on chronic occupational stress, mapped against the demand profile of a hospital switchboard role · prepared 22 August 2026

■ What this document is, and is not

This is **not a clinical assessment, not an expert report, and not evidence of anything that occurred inside Cory Shepherd's brain**. No neuroimaging, no cortisol assay, no biomarker of any kind has been obtained. Every neuroscience proposition below is drawn from the published literature on chronic occupational stress in general populations. It is written for Cory's own understanding and to help frame questions for his treating psychiatrist, Dr Krishnaiah, in the same channel already used for workplace-conditions material handed to him for job-demands context. **It must not be disclosed, served, filed, or put to the Commission or the Regulator in any form, and no diagnostic or causal conclusion may be drawn from it.** The diagnosis and the causal opinion in this matter are Dr Krishnaiah's independent clinical judgment alone.

Why it exists

The WHS cognitive-load analysis already prepared for this matter describes the *occupational* side of the picture well: a queue that rarely empties, frequent task-switching, emergency-code interruptions, and insufficient recovery between demands. This briefing adds the other half - what the general science says happens to a brain subjected to that kind of chronic demand pattern, over months rather than a single shift - so that the mechanism connecting workplace conditions to a psychological injury is something Cory can discuss intelligibly with his psychiatrist, rather than something asserted without foundation.

1. Allostatic load - the basic model

Allostasis is the process by which the body maintains stability by actively adjusting internal systems - cortisol, blood pressure, immune signalling, autonomic tone - in response to demand. **Allostatic load** is the cumulative wear that results when those adjustments are triggered repeatedly, or the system fails to switch off between demands (McEwen & Stellar, 1993; McEwen, 1998).

McEwen describes four conditions that generate allostatic load. The occupational profile in the WHS analysis maps onto at least three of them as a matter of general principle - this is a structural observation about the type of work, not a finding about Cory specifically:

McEwen's condition	General description	How the switchboard demand profile maps, as hypothesis
Repeated hits	The same stressor recurs often, without habituation	A queue that rarely reaches zero, over an entire shift, repeated across a working week
Failure to habituate	The stress response should quiet with repetition of a predictable stressor but does not	Emergency-code activation is high-consequence by design and is not the kind of stimulus the stress system habituates to
Failure to shut off	The response continues after the demand has passed	Interruption research describes a 'resumption lag' - unfinished task states persist in working memory after the interrupting demand ends

The fourth condition - inadequate response, where a system under-reacts and other systems compensate - is not addressed here, as it requires physiological data this briefing does not have and should not speculate about.

2. McEwen's model of stress-induced neural remodeling

A separate and more specific body of work, associated particularly with Bruce McEwen and colleagues, describes structural changes observed in animal models (and, with more variability, in human neuroimaging studies) after sustained stress exposure. Three regions recur in this literature:

Medial prefrontal cortex (mPFC)

The mPFC supports executive function: planning, working-memory manipulation, cognitive flexibility, and top-down regulation of the amygdala's threat response. Chronic stress models show **dendritic debranching** in this region - a

reduction in the complexity of dendritic branching on pyramidal neurons - associated in animal studies with reduced capacity for cognitive flexibility and for regulating emotional reactivity (Radley et al., 2004; Liston et al., 2006; Arnsten, 2009). Functionally, this is the region whose sustained engagement is demanded by the task-switching, working-memory holding, and inhibition-of-distraction load described in the WHS cognitive-load analysis.

Amygdala

The amygdala drives rapid threat detection and the emotional salience of a stimulus. Where the mPFC shows debranching under chronic stress, the amygdala in the same animal models shows the opposite pattern - **dendritic hypertrophy** (increased branching) in the basolateral nucleus, associated with heightened and more persistent reactivity to subsequent stressors (Vyas et al., 2002; Roozendaal, McEwen & Chattarji, 2009). The functional pairing matters: as the region that inhibits an overreaction becomes less complex, the region that generates the reaction becomes more reactive.

Hippocampus

The hippocampus supports contextual memory and exerts negative feedback on the hypothalamic-pituitary-adrenal (HPA) axis - it is part of what tells the stress response to switch off. Sustained glucocorticoid exposure is associated in the literature with dendritic atrophy in the CA3 region and, in some human cohort studies of chronic stress and PTSD, reduced hippocampal volume (Sapolsky, 1996; McEwen, 2000; Bremner, 1999). Reduced hippocampal regulation is one proposed mechanism by which a stress response, once triggered, becomes harder to switch off - consistent with the 'failure to shut off' allostatic-load condition described above.

■ The limits of this literature, stated plainly

These findings come overwhelmingly from animal models (typically rodent) and from human cohort studies using group-level neuroimaging, not from any assessment of a specific individual's brain. They describe a plausible, well-replicated general mechanism connecting sustained occupational stress to changes in cognitive and emotional regulation. **They do not establish, and this document does not assert, that any such change occurred in Cory Shepherd.** Whether his diagnosed condition involves these mechanisms is a question for his treating psychiatrist, informed by his clinical presentation - not something this briefing can determine.

3. Vulnerability matrix - mapping occupational demand to the stress-response literature

This matrix places the demand elements already identified in the WHS cognitive-load analysis alongside the general mechanism from the literature that a treating clinician might consider relevant. **It is a discussion tool, not a set of findings.** Every row states a general proposition from the literature next to an occupational feature; neither column says anything about Cory's actual physiology.

Occupational demand feature (from the WHS analysis)	General allostatic-load concept	Region implicated in the general literature
Queue rarely reaches zero; little interval for attention to reset	Repeated hits without recovery interval; failure to habituate	mPFC (sustained executive load)
Frequent task-switching between call, directory, paging/code system, and unfinished prior tasks	Working-memory occupation by unfinished task states ('resumption lag')	mPFC (working-memory / cognitive flexibility)
Emergency-code activation: high urgency, high accuracy demand, concurrent with queued calls	Acute, high-salience threat-type stimulus, repeated over a working week	Amygdala (threat salience and reactivity)
Exposure to caller distress, complaint, blame or hostility	Chronic exposure to socially/emotionally threatening stimuli	Amygdala; mPFC-amygdala regulatory coupling
Insufficient recovery between shifts; night or rotating work; on-call ambiguity	Failure of the stress response to switch off between demand periods	Hippocampus (HPA-axis negative feedback)
Sustained period of the above across months, not a single shift	Cumulative allostatic load rather than a single acute stressor	All three regions, cumulatively, per the literature

The final row is the one worth raising directly with Dr Krishnaiah: the WHS material describes a demand pattern sustained over roughly a year (the pleaded period in the Amended Statement of Facts and Contentions runs from mid-2023), not an isolated bad shift. The allostatic-load literature is specifically about cumulative exposure over that kind of timeframe, which is the reason it may be a useful frame for him to consider rather than a reason to draw any conclusion from it here.

4. General recovery considerations - not a treatment plan

■ Not medical advice

This section restates general, widely published findings about what supports recovery from chronic stress exposure. It is not a treatment plan, is not tailored to Cory's diagnosis or presentation, and must not substitute for anything Dr Krishnaiah or his GP actually prescribes or recommends. Anything in this section should be raised with them, not acted on independently where it might conflict with clinical advice already given.

Restoring HPA-axis regulation

The literature on hippocampal-mediated feedback suggests that regularising sleep-wake timing, reducing unpredictable demand, and consistent daily structure support the systems that switch a stress response off. This is consistent with why removal from the triggering environment - as has occurred here since June 2024 - is itself considered by the literature to be part of recovery, not merely an accommodation.

Working-memory and executive recovery

mPFC-dependent function is described in the literature as recovering, at least partially, once chronic demand is removed (Liston et al., 2009, found partial reversal of stress-induced dendritic changes after a recovery period in animal models). This is consistent with cognitive symptoms - difficulty concentrating, word-finding, decision fatigue - being expected to improve gradually with sustained removal from the stressor, rather than resolving immediately.

Amygdala reactivity

Reduction in generalised threat-reactivity is described in the literature as the slower-recovering domain, and as the one most responsive to structured therapeutic intervention (rather than simple removal of the stressor) - relevant to why psychiatric or psychological treatment, not time alone, is usually the indicated pathway for this component.

Structure and reintegration

The literature on return-to-work after stress-related psychological injury generally treats graduated, structured re-engagement - rather than either full removal or full resumption - as the pathway associated with the best functional outcome, which is a matter for Dr Krishnaiah's own assessment of timing and capacity.

Questions this raises for the appointment on 27 August 2026

Framed as questions for Dr Krishnaiah to consider, not propositions for him to adopt:

- Does the sustained, cumulative nature of the workplace demand (as opposed to a single incident) fit the clinical picture he has observed - in particular, whether the presentation looks more like response to a discrete event or to prolonged cumulative load?
- Do the cognitive symptoms reported (concentration, word-finding, decision fatigue) fit a pattern consistent with prefrontal-mediated executive difficulty, and does that inform his view on functional capacity and timing of return to work?
- Does the general framework of cumulative allostatic load - as opposed to a single-stressor model - assist him in articulating causation and mechanism in his report, to the extent he considers it clinically appropriate to do so?

Sources referred to

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These are landmark and widely cited papers in the stress-neuroscience literature, reproduced here from general knowledge of the field. They have not been independently re-verified against original text for this briefing and should be checked before any citation is relied upon for a purpose beyond framing a conversation.